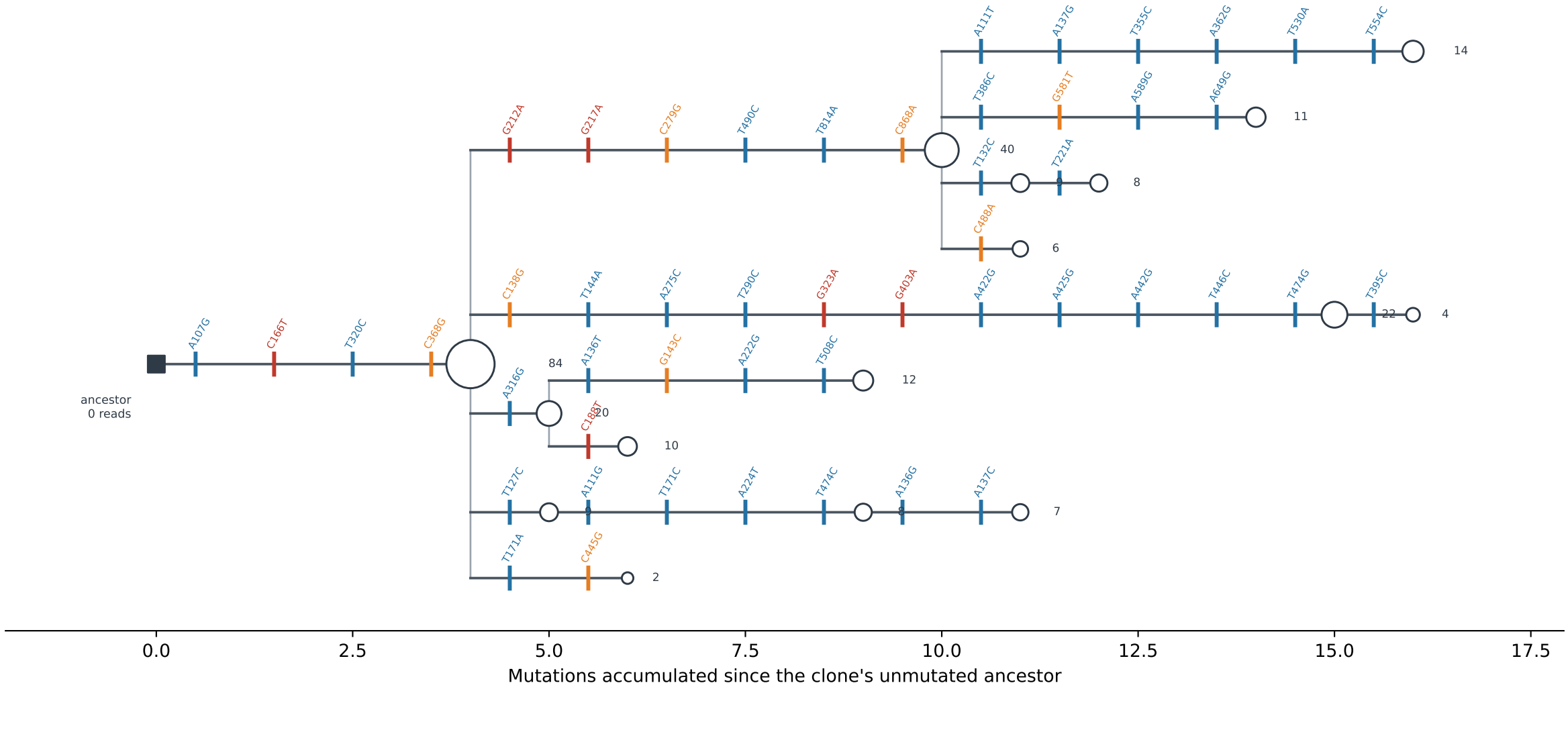
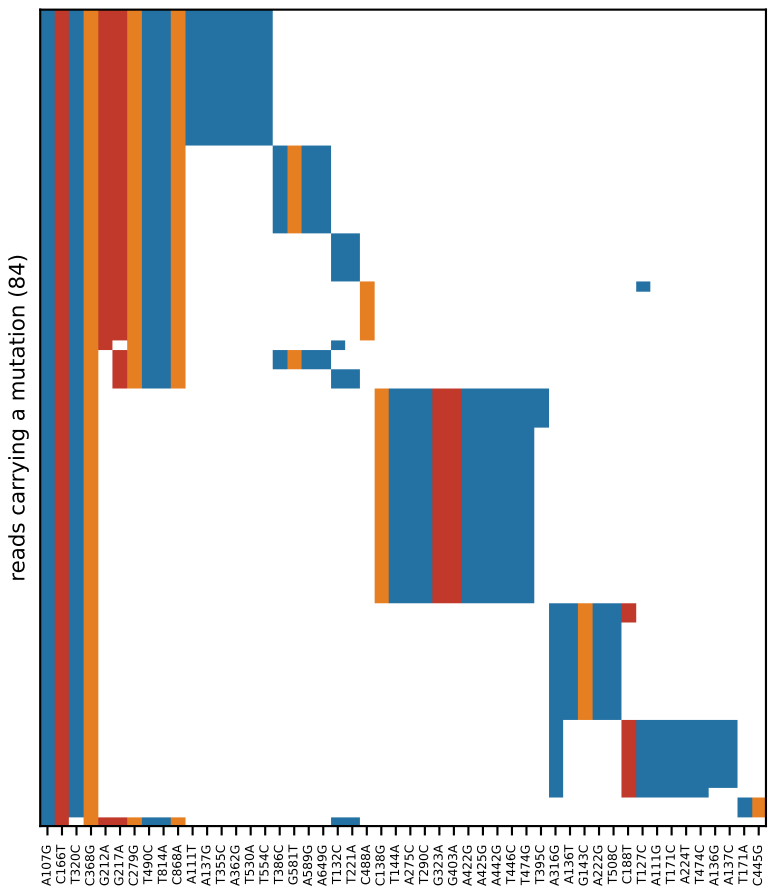
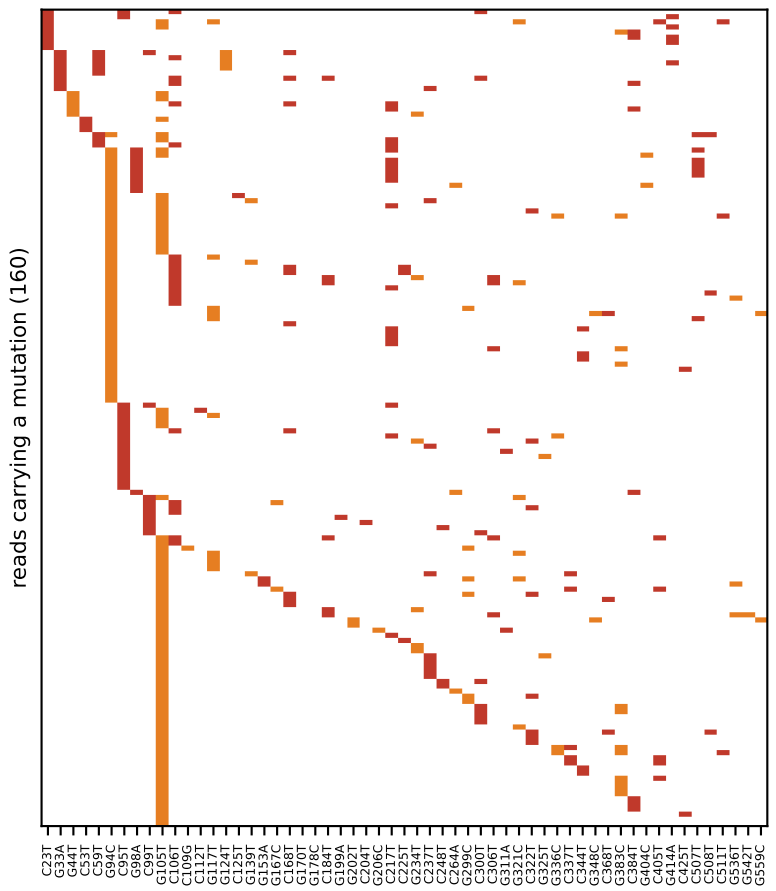
**A** $J_H4$ **intron, Wt3 clone in vivo · 84 reads · 87% of mutated reads fit one tree**



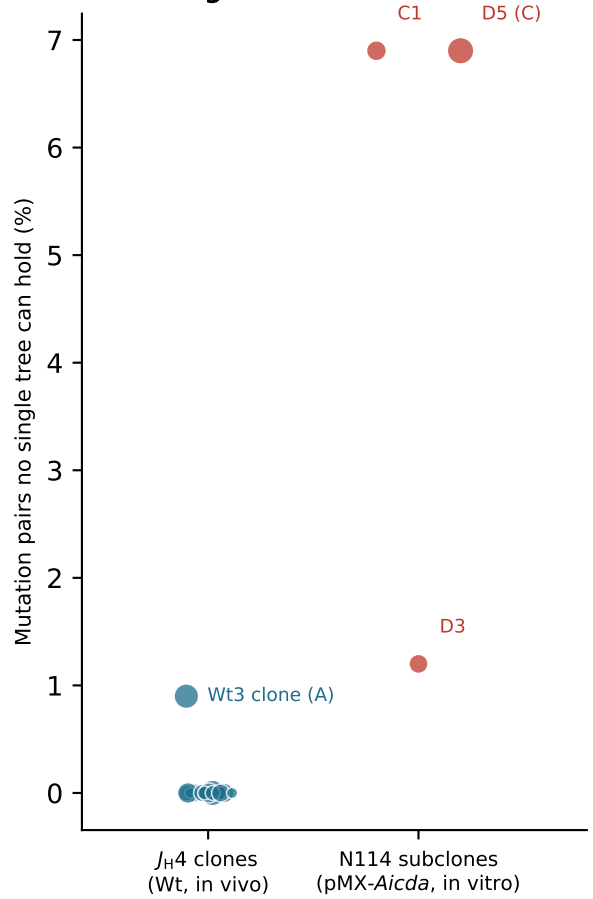

**B** **Wt3 clone: nested**




**C** **N114 subclone D5, in vitro: not nested**




**D** **Four-gamete test**




G:C transition  G:C transversion  A:T mutation